Dataset Information Overview

BacTools is designed to catalog computational tools for bacterial data analysis by using user generated keywords to extract metadata from public software repositories. The primary data consists of software metadata records from CRAN, Bioconductor, and GitHub, with a special focus on tools relevant to bacterial genomics, proteomics, and microbiome research. This metadata is standardized and organized for downstream comparison, filtering, and selection by researchers.

Input:
User- submitted keyword (e.g., "bacteria," "microbiome," "genomics," "16S," "antibiotic").

Data Format

- Keyword in string format (case, dash insensitive)

Data Preprocessing Steps

- Expand search to include case insensitive and regex versions of search terms for queries (microbiome, microbiomes, MICROBIOME)
- Tag/classify tools by application (e.g., metagenomics, phylogenetics)
- Handle missing and inconsistent metadata fields
- Harmonize data fields across platforms
- Remove duplicate tools (merged by name/DOI) and export final list to csv

Example Metadata Subset (for Development)

A representative subset comprises:

- One keyword ("microbiome")

Example csv table output:

name	author	summary	doi	citation
CoreMicrobiomeR	Sorna A M [aut], Mohammad Samir Farooqi [aut, cre], Dwijesh Chandra Mishra [aut], Krishna Kumar Chaturvedi [aut], Anu Sharma [aut], Prawin Arya [aut], Sudhir Srivastava [aut],	Identification of Core Microbiome	10.32614/CRAN.package.CoreMicrobiomeR	CoreMicrobiomeR. (2024). Identification of Core Microbiome. R package version 0.1.0. https://doi.org/10.32614/CRAN.package.CoreMicrobiomeR

	Sharanbasappa [aut], Girish Kumar Jha [aut], Kabilan S [ctb]			
metamicrobiomeR	Nhan Ho [aut, cre]	Microbiome Data Analysis & Meta-Analysis with GAMLSS-BEZI & Random Effects	10.32614/CRAN.package.metamicrobiomeR	metamicrobiomeR. (2020). Microbiome Data Analysis & Meta-Analysis with GAMLSS-BEZI & Random Effects. R package version 1.2. https://doi.org/10.32614/CRAN.package.metamicrobiomeR